

Week 2 — solutions

September 18, 2026

Exercise 1. Which of the following represents the CNF and DNF forms of the compound proposition:

$$(p \vee \neg q) \wedge (\neg p \vee q)$$

☐ CNF: $(p \vee \neg q) \wedge (\neg p \vee q)$, DNF: $(p \vee \neg q) \wedge (\neg p \vee q)$

☒ CNF: $(p \vee \neg q) \wedge (\neg p \vee q)$, DNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$

☐ CNF: $(p \vee q) \wedge (\neg p \vee \neg q)$, DNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$

☐ CNF: $(p \wedge q) \vee (\neg p \wedge \neg q)$, DNF: $(p \vee q) \wedge (\neg p \vee \neg q)$

Given the proposition $(p \vee \neg q) \wedge (\neg p \vee q)$:

- It is already in CNF as it's a conjunction of two clauses.
- For DNF, we first apply the distributive law and get:

$$\begin{aligned} & (p \vee \neg q) \wedge (\neg p \vee q) \\ \equiv & [(p \vee \neg q) \wedge (\neg p)] \vee [(p \vee \neg q) \wedge (q)] \end{aligned}$$

We apply the distributive law again and get:

$$\begin{aligned} & [(p \vee \neg q) \wedge (\neg p)] \vee [(p \vee \neg q) \wedge (q)] \\ \equiv & \left[\underbrace{(p \wedge \neg p)}_{\equiv F} \vee (\neg q \wedge \neg p) \right] \vee \left[(p \wedge q) \vee \underbrace{(\neg q \wedge q)}_{\equiv F} \right] \\ \equiv & (\neg q \wedge \neg p) \vee (p \wedge q) \end{aligned}$$

This gives the CNF as $(p \vee \neg q) \wedge (\neg p \vee q)$ and the DNF as $(p \wedge q) \vee (\neg p \wedge \neg q)$. Thus, the correct option is the second one.

Exercise 2.

1. Find a DNF of $p \oplus (q \wedge r)$ without using a truth table.

Recall that $a \oplus b \equiv (a \wedge \neg b) \vee (\neg a \wedge b)$.

Using this, the given proposition $p \oplus (q \wedge r)$ can be rewritten as:

$$\begin{aligned} & (p \wedge \neg(q \wedge r)) \vee (\neg p \wedge (q \wedge r)) \\ \equiv & (p \wedge (\neg q \vee \neg r)) \vee (\neg p \wedge (q \wedge r)) \text{ De Morgan's Law} \\ \equiv & ((p \wedge \neg q) \vee (p \wedge \neg r)) \vee (\neg p \wedge (q \wedge r)) \text{ Distributive Law} \\ \equiv & (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge q \wedge r) \end{aligned}$$

This is a disjunction of conjunctions, which means it's in DNF.

$$\text{DNF: } (p \wedge \neg q) \vee (p \wedge \neg r) \vee (\neg p \wedge q \wedge r)$$

2. Find a CNF of $[(p \rightarrow q) \vee (r \rightarrow s)] \wedge u$ without using a truth table.

Firstly, we rewrite $(p \rightarrow q) \vee (r \rightarrow s)$ using $a \rightarrow b \equiv \neg a \vee b$:

$$(\neg p \vee q) \vee (\neg r \vee s)$$

Which is equivalent to:

$$\neg p \vee q \vee \neg r \vee s$$

By replacing the in original expression, we get

$$[\neg p \vee q \vee \neg r \vee s] \wedge u$$

Which is a CNF.

Exercise 3. (Warm-up; unlike the multiple choice questions at the final exam, more than a single answer may be correct.). The law regarding alcohol drinking in Switzerland states the following: “If you are drinking strong alcohol, then you must be at least 18 years old.” (for all x , $A(x) \rightarrow 18(x)$). Assume that you are hired as a police officer at Satellite and need to enforce the above law. Which student(s) are you going to arrest and why:

- ☐ The 17-year old student drinking soda.
- ☒ The 17-year old student drinking vodka.
- ☐ The 18-year old student drinking water.
- ☐ The 18-year old student drinking tequila.

Next Tuesday, Satellite organizes a cabaret night open to all students under the following conditions: “Everyone must be at least 18 years old and drink strong alcohol” (for all x , $A(x) \wedge 18(x)$). As previously, which student(s) are you going to arrest and why:

- ☒ The 17-year old student drinking soda.
- ☒ The 17-year old student drinking vodka.
- ☒ The 18-year old student drinking water.
- ☐ The 18-year old student drinking tequila.

Exercise 4. (Rosen, exercise 8, 1.4.14 in 8th edition) Translate these statements into English, where $R(x)$ is “ x is a rabbit” and $H(x)$ is “ x hops” and the domain consists of all animals.

1. $\forall x(R(x) \rightarrow H(x))$: All rabbits hop.
2. $\exists x(R(x) \rightarrow H(x))$: Multiple possible solutions (they are all correct):
 - There is an animal that hops if it's a rabbit
 - There exists an animal where if it is a rabbit, then it hops
 - There exists an animal such that if it is a rabbit, then it hops

3. $\forall x(R(x) \wedge H(x))$: All animals are rabbits and also hop.
4. $\exists x(R(x) \wedge H(x))$: There is a rabbit that also hops.

Exercise 5.

1. Which expressions below are equivalent to $\neg(\forall x \exists y P(x, y))$. Explain.

- ✓ $\exists x \forall y \neg P(x, y)$;
 ○ $\exists x \exists y \neg P(x, y)$.

The formula

$$\forall x \exists y P(x, y)$$

states that for every x , there exists some y such that $P(x, y)$ is true.

We find its negation by moving the negation inward step by step:

$$\neg(\forall x \exists y P(x, y)) \leftrightarrow \exists x \neg(\exists y P(x, y)) \leftrightarrow \exists x \forall y \neg P(x, y).$$

In words, the negation says that there exists some x for which no y makes $P(x, y)$ true, i.e., $P(x, y)$ is false for every y .

Therefore, the first expression is equivalent. The second is not, since it only requires $P(x, y)$ to fail for one pair (x, y) .

Exercise 6. (Rosen, exercise 9, 1.5.8 in 8th edition) Let $L(x, y)$ be the statement “ x loves y ”, where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements:

1. Everybody loves Sharon.

$$\forall x L(x, \text{Sharon}).$$

2. Everybody loves somebody.

$$\forall x \exists y L(x, y).$$

3. There is somebody whom everybody loves.

$$\exists x \forall y L(y, x).$$

4. Nobody loves everybody.

$$\forall x \exists y \neg L(x, y).$$

5. There is somebody whom Daisy does not love.

$$\exists x \neg L(\text{Daisy}, x).$$

6. There is somebody whom no one loves.

$$\exists x \forall y \neg L(y, x).$$

7. There is exactly one person whom everybody loves.

$$\exists x (\forall y L(y, x) \wedge \forall z ((\forall w L(w, z)) \rightarrow z = x)).$$

8. There are exactly two people whom Marsellus loves.

$$\exists x \exists y (x \neq y \wedge L(\text{Marsellus}, x) \wedge L(\text{Marsellus}, y) \wedge \forall z (L(\text{Marsellus}, z) \rightarrow (z = x \vee z = y))).$$

9. *Everyone loves himself or herself.*

$$\forall x L(x, x).$$

10. *There is someone who loves no one besides himself or herself.*

There are two ways to interpret this question:

A) To interpret it as "there is someone who doesn't love anyone who is not himself/herself, but may or may not love himself/herself," in which case, the solution is:

$$\exists x \forall y (L(x, y) \rightarrow x = y).$$

B) To interpret it as "there is someone who doesn't love anyone who is not himself/herself, but loves himself/herself," in which case, the solution is:

$$\exists x \forall y (L(x, x) \wedge (L(x, y) \rightarrow x = y)).$$

Exercise 7. *Given the two statements below, where the domain of discourse is $\mathbf{R}$ for both x and y ,*

$$\exists y \forall x (x \neq 0 \rightarrow xy = 1)$$

$$\exists x \forall y (xy < 0 \rightarrow xy > 0)$$

- ☐ They are both false.
- ☐ Only the first is true.
- ☒ Only the second is true.
- ☐ They are both true.

The first statement is "obviously incorrect" because not all non-zero x -values can be the inverse of some particular y -value. But let's do it more carefully: if the first statement is False, then its negation must be True. So, consider the negation of the first statement:

$$\begin{aligned} \neg(\exists y \forall x (x \neq 0 \rightarrow xy = 1)) &\equiv \neg(\exists y \forall x (\neg(x \neq 0) \vee xy = 1)) \\ &\equiv \forall y \exists x (x \neq 0 \wedge \neg(xy = 1)) \\ &\equiv \forall y \exists x (x \neq 0 \wedge xy \neq 1). \end{aligned}$$

We have to show that this final statement " $\forall y \exists x (x \neq 0 \wedge xy \neq 1)$ " is True: for $y = 0$ one can take any x with $x \neq 0$ (because $xy = 0$ and thus $xy \neq 1$), and for $y \neq 0$ one can take $x = \frac{2}{y}$ (because then $x \neq 0$ and $xy = 2$ and thus $xy \neq 1$). So, irrespective of the value of y , a non-zero x -value can be found such that xy is not equal to one, proving that indeed " $\forall y \exists x (x \neq 0 \wedge xy \neq 1)$ " is True. Because the negation of the first statement is True, the first statement is False.

For the statement " $\exists x \forall y (xy < 0 \rightarrow xy > 0)$ ", consider $x = 0$, then for all y it is the case that $xy < 0$ is False and thus " $xy < 0 \rightarrow xy > 0$ " is True. It follows that an x -value exists such that for all y -values the statement " $xy < 0 \rightarrow xy > 0$ " is True: thus the second statement is True.

Exercise 8.

Consider the two statements below, where $P(x, y)$ is a propositional function and the domain of discourse is $\mathbf{Z}_{\geq 0}$ for x, y and z :

$$(\exists y \forall x P(x, y)) \rightarrow (\forall x \exists y P(x, y))$$

$$(\neg \exists x x^x = x!) \rightarrow \forall y, z y \neq z.$$

- ☐ They are both false.
- ☐ Only the first is true.

○ Only the second is true.

✓ They are both true.

If $\exists y \forall x P(x, y)$ is True, there is some value $\tilde{y}$ (formally: use existential instantiation) such that $\forall x P(x, \tilde{y})$ is True. Thus (formally using existential generalization) $\forall x \exists y P(x, y)$. It follows that the first statement is True.

The statement $\neg \exists x x^x = x!$ is equivalent to $\forall x x^x \neq x!$. Because $x = 1$ is in the domain $\mathbf{Z}_{\geq 0}$ for x it follows from $\forall x x^x \neq x!$ (formally: use universal instantiation) that $1^1 \neq 1!$; but $1^1 = 1 = 1!$ so $\forall x x^x \neq x!$ is False. Because the statement “False $\rightarrow$ q ” is True for any (logical) value q , it follows that the second statement is True.

Exercise 9. Let E be a set of endpoints on a network, let P be a set of paths connecting those endpoints, and let $C(p, x, y)$ be the proposition that path $p \in P$ connects endpoints x and y with $x, y \in E$. The statement “there are at least two paths connecting every two distinct endpoints on the network” can be expressed by

○ $\forall x, y \in E \left(x \neq y \rightarrow \exists p, q \in P (p \neq q \wedge (C(p, x, y) \vee C(q, x, y))) \right)$.

○ $\forall x, y \in E \left(x \neq y \wedge \exists p, q \in P (p \neq q \wedge C(p, x, y) \wedge C(q, x, y)) \right)$.

✓ $\neg \left(\exists x, y \in E (x \neq y \wedge \forall p, q \in P (p = q \vee \neg C(p, x, y) \vee \neg C(q, x, y))) \right)$.

○ $\neg \left(\exists x, y \in E (x \neq y \wedge \forall p, q \in P (p = q \wedge \neg C(p, x, y) \wedge \neg C(q, x, y))) \right)$.

Given any two distinct endpoints (i.e., $\forall x, y \in E$ if $x \neq y$ then ...) there are at least two paths (i.e., $(x \neq y \rightarrow \exists p, q \in P (p \neq q \wedge C(p, x, y) \wedge C(q, x, y)))$), leading to the complete logical expression

$$\forall x, y \in E (x \neq y \rightarrow \exists p, q \in P (p \neq q \wedge C(p, x, y) \wedge C(q, x, y))).$$

This is equivalent to

$$\neg \neg \left(\forall x, y \in E (\neg(x \neq y) \vee \exists p, q \in P (p \neq q \wedge C(p, x, y) \wedge C(q, x, y))) \right)$$

and thus to

$$\neg \left(\exists x, y \in E (x \neq y \wedge \neg \exists p, q \in P (p \neq q \wedge C(p, x, y) \wedge C(q, x, y))) \right)$$

and finally to

$$\neg \left(\exists x, y \in E (x \neq y \wedge \forall p, q \in P (p = q \vee \neg C(p, x, y) \vee \neg C(q, x, y))) \right).$$

The other three possibilities can be seen to be wrong in various different ways.

Exercise 10. Given the propositional function $T(x)$, the statement $\exists! x T(x)$ is logically equivalent to

✓ $\neg(\forall x [T(x) \rightarrow \exists y \neq x T(y)])$.

○ $\exists x \forall y ((\neg T(y)) \vee (y = x))$.

○ $\exists x (T(x) \vee \forall y ((\neg T(y)) \vee (y = x)))$.

○ $\exists x (T(x) \wedge \forall y [T(y) \wedge (y = x)])$.

The second and third statements are True for the propositional function $T(x)$ with a non-empty domain that is False for all x (and for which the statement $\exists! x T(x)$ is thus False). The fourth statement is False for, for instance, the propositional function $T(x)$ that consists of the statement “ x equals 1” and where the domain of x contains the element 1 and at least one other element, whereas for that same propositional function the statement $\exists! x T(x)$ is True.

This leaves only the first statement, and indeed it is equivalent to $\exists! x T(x)$ because $\neg(\forall x [T(x) \rightarrow \exists y \neq x T(y)])$ is equivalent to $\neg(\forall x [\neg T(x) \vee \exists y \neq x T(y)])$ and thus equivalent to $\exists x T(x) \wedge \forall y \neq x \neg T(y)$.

Exercise 11. Given the propositional functions $G(x)$: “ x is a boy”, $F(y)$: “ y is a girl”, and $A(z)$: “ z likes computers”, the statement “all boys like computers and there is a girl that does not like computers” can be expressed by

- ✓ $\neg[(\exists y (G(y) \wedge \neg A(y))) \vee (\forall x (F(x) \rightarrow A(x)))]$.
- $\neg[(\forall x (F(x) \rightarrow A(x))) \vee (\exists y (G(y) \rightarrow \neg A(y)))]$.
- $(\exists x (F(x) \rightarrow \neg A(x))) \wedge (\forall y (\neg G(y) \vee A(y)))$.
- $(\forall y (G(y) \wedge A(y))) \wedge (\exists x (F(x) \wedge \neg A(x)))$.

Using the equivalence $p \rightarrow q \equiv \neg p \vee q$, the first answer is equivalent to $\neg[(\exists y (G(y) \wedge \neg A(y))) \vee (\forall x (\neg F(x) \vee A(x)))]$, implying it is equivalent to $(\forall y (\neg G(y) \vee A(y))) \wedge (\exists x (F(x) \wedge \neg A(x)))$ and thus to $(\forall y (G(y) \rightarrow A(y))) \wedge (\exists x (F(x) \wedge \neg A(x)))$. This says that for all elements of the domain it is the case that if the element is a boy, then that boy like computers (“ $\forall y (G(y) \rightarrow A(y))$ ”) and that furthermore (“ $\wedge$ ”) there exists an element of the domain that is a girl that does not like computers (“ $\exists x (F(x) \wedge \neg A(x))$ ”). This corresponds to the statement “all boys like computers and there is a girl that does not like computers”. Note that this statement does not imply that there are any boys in the domain – but if there are boys, then those boys like computers.

The other answers are incorrect:

- Twice using the same equivalence $p \rightarrow q \equiv \neg p \vee q$ again, the second answer is equivalent to $\neg[(\forall x (\neg F(x) \vee A(x))) \vee (\exists y \neg(G(y) \vee \neg A(y)))]$, implying it is equivalent to $(\exists x (F(x) \wedge \neg A(x))) \wedge (\forall y (G(y) \wedge A(y)))$. This says that there exists an element of the domain that is a girl that does not like computers (“ $\exists x (F(x) \wedge \neg A(x))$ ”) and that furthermore (“ $\wedge$ ”) all elements of the domain are boys that like computers (“ $\forall y (G(y) \wedge A(y))$ ”): this is contradictory because the girl (that does not like computers) cannot at the same time be one of the boys (that like computers).
- The third answer says that there exists an element of the domain such that if that element is a girl then that girl does not like computers (“ $\exists x (F(x) \rightarrow \neg A(x))$ ”) combined with (“ $\wedge$ ”) a condition that does not express anything about “girls”. It follows that the first part is True if there are no girls in the (non-empty) domain, whereas the original statement says that “there is a girl ...”.
- The fourth answer is equivalent to the reformulated version of the second answer (as derived above) because $p \wedge q \equiv q \wedge p$.